# Area ratios without lengths

L2-X1 Level 2 Intermediate · Geometry · Beyond the textbook

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November 2026

Areas as unknowns and ratios on lines as equations: same-height triangles, cevians, trapezia and parallelograms, and the triangle formed inside by three cevians.
**By the end you can** turn a ratio on a line into a ratio of areas, and back, using a common apex or a common base; find unknown areas in a triangle cut by cevians by labelling them and solving linear equations; use the area ratios of a trapezium cut by its diagonals, and find the triangle formed inside by three cevians.
**Prerequisites:** Area, Pythagoras and similar shapes (L2-GE2). **Contests:** IMC, Cayley, Hamilton, Intermediate Kangaroo, AMC 10. **Time:** about 75 minutes.

## 1 Lesson

*Triangle $ABC$ has area 30. The point $D$ lies on $BC$ with $BD = 2DC$, and $E$ is the midpoint of $AD$. Find the area of triangle $BEC$.* Triangles $ABD$ and $ADC$ have the same height from $A$, so their areas are in the ratio $BD : DC = 2 : 1$, giving $[ABD] = 20$ and $[ADC] = 10$. Since $E$ is the midpoint of $AD$, the same argument with apex $B$ gives $[BED] = \frac{1}{2}[ABD] = 10$, and with apex $C$ gives $[CED] = 5$. Hence $[BEC] = 15$. No length in the problem is known, and none was needed.

Here $[XYZ]$ denotes the area of triangle $XYZ$. The fact used above is Proposition 1 of Book VI of Euclid's *Elements*: triangles under the same height are to one another as their bases. School courses use area as $\frac{1}{2} \times$ base $\times$ height with numbers for the lengths, while contest problems often give only ratios, or only areas, and ask for another ratio or area. The *area method* treats areas as the unknowns and ratios on lines as the equations. It is a standard topic in Chinese olympiad training, and Zhang Jingzhong developed it into a complete method that was later used for computer proofs (Chou, Gao and Zhang, *Machine Proofs in Geometry*, 1994). We assume L2-GE2: areas by decomposition and similar triangles.

### 1.1 1. Four facts

**Fact 1 (same height).**
If $X$, $Y$, $Z$, $W$ lie on one line and $P$ does not, then $[PXY] : [PZW] = XY : ZW$. In particular, a cevian $AD$ of triangle $ABC$ gives $[ABD] : [ADC] = BD : DC$.
**Fact 2 (cevian ratio).** If $P$ is a point of the cevian $AD$ other than $A$, then $[ABP] : [ACP] = BD : DC$.
**Fact 3 (common base).** If the segments $AD$ and $XY$ meet at $P$, then $[AXY] : [DXY] = AP : PD$.
**Fact 4 (common angle).** If $D$ lies on $AB$ and $E$ on $AC$, then $\dfrac{[ADE]}{[ABC]} = \dfrac{AD}{AB} \cdot \dfrac{AE}{AC}$.

*Proof.* s. Fact 1: both triangles have the distance from $P$ to the line as height. Fact 2: let $BD : DC = r$. By Fact 1 with apex $A$, $[ABD] = r[ACD]$, and with apex $P$, $[PBD] = r[PCD]$. Subtracting,



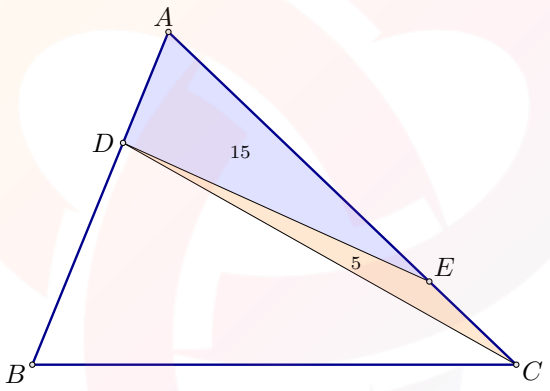
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$[ABP] = [ABD] - [PBD] = r([ACD] - [PCD]) = r[ACP]$. Fact 3: drop perpendiculars $AH$ and $DK$ to line $XY$. The right-angled triangles $AHP$ and $DKP$ have equal angles at $P$, so they are similar and $AH : DK = AP : PD$; these are the heights of $AXY$ and $DXY$ on the common base $XY$. Fact 4: by Fact 1, $[ADE] = \frac{AD}{AB}[ABE]$ and $[ABE] = \frac{AE}{AC}[ABC]$. ■

Fact 2 is the one most students do not know, and it is the one that sets up the equations in Section 3: a single ratio on a line is shared by two pairs of triangles, one pair with apex $A$ and one with apex $P$. Fact 3 gives the ratio in which a cevian is cut by another line, which is what is needed for the area of the triangle formed inside by three cevians (Example 4).

**Example 1.**
Triangle $ABC$ has area 60. The point $D$ lies on $AB$ with $AD : DB = 1 : 2$, and $E$ lies on $AC$ with $AE : EC = 3 : 1$. Find the areas of triangles $ADE$ and $DEC$.

*Solution.*
By Fact 4, $[ADE] = \frac{1}{3} \cdot \frac{3}{4} \cdot 60 = 15$. For $DEC$ we use apex $D$ and the line $AC$: by Fact 1, $[DEC] : [DAE] = EC : AE = 1 : 3$, so $[DEC] = 5$. As a check, $[ADC] = \frac{1}{3} \cdot 60 = 20 = 15 + 5$.




## 1.2 2. Trapezia and parallelograms

**Fact 5 (trapezium).**
Let $ABCD$ be a trapezium with $AB \parallel DC$ and $AB = k \cdot DC$, whose diagonals meet at $X$. Then

$$[ABX] : [BCX] : [CDX] : [DAX] = k^2 : k : 1 : k.$$

In any convex quadrilateral with diagonals meeting at $X$, $[ABX] \cdot [CDX] = [BCX] \cdot [DAX]$.

*Proof.* For any convex quadrilateral, Fact 1 with apex $B$ and then apex $D$ gives $[ABX] : [BCX] = AX : XC = [DAX] : [CDX]$, which is the second statement. In the trapezium, triangles $ABX$ and $CDX$ are similar (alternate angles), with ratio $AB : CD = k$, so $AX : XC = k$ and $[ABX] = k^2[CDX]$. Also $[ABD] = [ABC]$, since they share the base $AB$ and $D$, $C$ are at the same distance from it; removing $[ABX]$ from both gives $[DAX] = [BCX]$. Finally $[BCX] = \frac{XC}{AX}[ABX] = \frac{1}{k} \cdot k^2[CDX] = k[CDX]$. ■

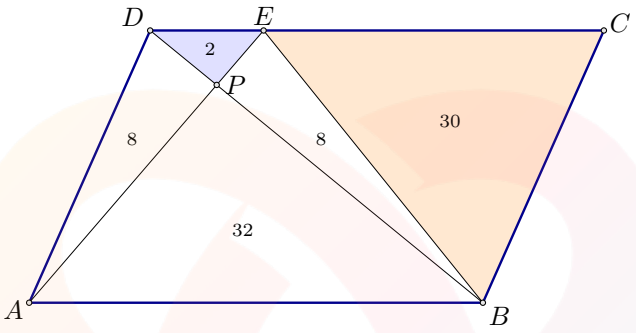
A parallelogram is cut by a diagonal into two halves, and any triangle with one side of the parallelogram as base and its third vertex on the opposite side has half the area. A segment from a vertex to a point of a non-adjacent side creates a trapezium, and Fact 5 applies to it.



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**Example 2.**
Parallelogram $ABCD$ has area 80. The point $E$ lies on $CD$ with $DE : EC = 1 : 3$, and $AE$ meets the diagonal $BD$ at $P$. Find the area of triangle $DPE$ and of quadrilateral $PBCE$.

*Solution.*
Since $AB \parallel DE$, the quadrilateral $ABED$ is a trapezium whose diagonals $AE$ and $BD$ meet at $P$, and $AB = 4\,DE$. By Fact 5 the triangles $ABP$, $BEP$, $EDP$, $DAP$ have areas in the ratio $16 : 4 : 1 : 4$, so $[ABED]$ is 25 parts. Triangle $BCE$ has base $EC = \frac{3}{4}DC$ and the height of the parallelogram, so $[BCE] = \frac{3}{4} \cdot 40 = 30$ and $[ABED] = 50$. One part is 2. Hence $[DPE] = 2$ and $[PBCE] = [BPE] + [BCE] = 8 + 30 = 38$.
The similar triangles $DPE$ and $BPA$ give $DP : PB = 1 : 4$ directly; the trapezium gives all four pieces at once.




## 1.3 3. Labelling the unknown areas

When two cevians $BE$ and $CF$ meet at $P$, the triangle is cut into four pieces, and drawing $AP$ makes five triangles. The method is as follows.

1. Draw the segment from the remaining vertex to the intersection point, so that every region is a triangle.
2. Write a letter for each unknown area.
3. For each cevian through $P$, read the ratio in which $P$ divides it from a pair of known triangles (Fact 1), and write the same ratio for the pair of triangles with apex at the third vertex (Fact 2 applied to that vertex). Each cevian gives one equation.
4. Solve the linear equations.

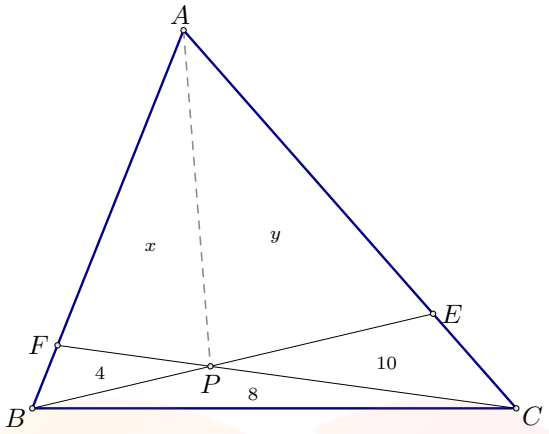
**Example 3.**
The cevians $BE$ and $CF$ of triangle $ABC$ meet at $P$. The areas of triangles $BPF$, $BPC$ and $CPE$ are 4, 8 and 10. Find the area of quadrilateral $AFPE$.

*Solution.*
No length is known, and a coordinate solution would have to choose a shape for the triangle with six unknown coordinates. Draw $AP$ and let $x = [APF]$, $y = [APE]$. Two unknowns, so we need two equations, one from each cevian.
On $CF$: triangles $BPF$ and $BPC$ have apex $B$, so $FP : PC = 4 : 8 = 1 : 2$. Triangles $APF$ and $APC$ have apex $A$ and the same bases, so $x : (y + 10) = 1 : 2$. On $BE$: triangles $BPC$ and $CPE$ have apex $C$, so $BP : PE = 8 : 10 = 4 : 5$, and with apex $A$, $(x + 4) : y = 4 : 5$.



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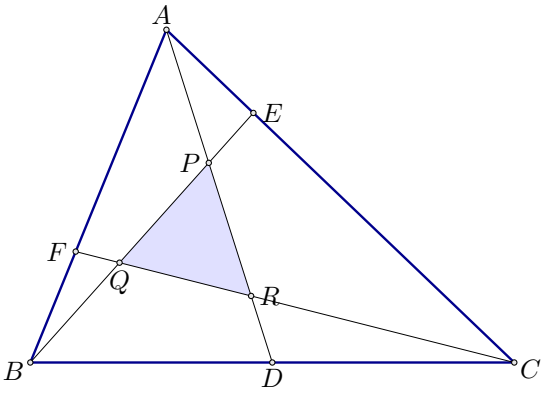
The equations are $2x = y + 10$ and $5x + 20 = 4y$. Substituting $y = 2x - 10$ gives $5x + 20 = 8x - 40$, so $x = 20$, $y = 30$ and $[AFPE] = 50$. The whole triangle has area 72, and you can check that $AF : FB = [AFC] : [BFC] = 60 : 12 = 5 : 1$.




**Example 4.**
In triangle $ABC$ of area 30, $D$ is the midpoint of $BC$, $E$ lies on $CA$ with $CE : EA = 3 : 1$, and $F$ lies on $AB$ with $AF : FB = 2 : 1$. Let $AD$ meet $BE$ at $P$, $BE$ meet $CF$ at $Q$, and $CF$ meet $AD$ at $R$. Find the area of triangle $PQR$.

*Solution.*
First the standard way. Area ratios do not depend on the shape of the triangle (a shear keeps every area, and a stretch in one direction multiplies every area by the same factor), so we could take $A = (0,0)$, $B = (3,0)$, $C = (0,20)$, find $P$, $Q$, $R$ by solving three pairs of linear equations, and finish with the shoelace formula. This works and takes a page of fractions.
With areas, note that the triangles $ABP$, $BCQ$, $CAR$ and $PQR$ fill triangle $ABC$ without overlapping, so $[PQR] = 30 - [ABP] - [BCQ] - [CAR]$. Each of the three corner triangles needs one ratio along a cevian, and Fact 3 gives it.
For $P$, use the common base $BE$: $AP : PD = [ABE] : [DBE]$. Now $[ABE] = \frac{1}{4} \cdot 30 = \frac{15}{2}$ and $[CBE] = \frac{45}{2}$, so $[DBE] = \frac{1}{2}[CBE] = \frac{45}{4}$. Hence $AP : PD = 2 : 3$ and $[ABP] = \frac{2}{5}[ABD] = \frac{2}{5} \cdot 15 = 6$.
For $Q$, use the base $CF$: $[BCF] = \frac{1}{3} \cdot 30 = 10$, $[ACF] = 20$ and $[ECF] = \frac{3}{4} \cdot 20 = 15$, so $BQ : QE = 10 : 15 = 2 : 3$ and $[BCQ] = \frac{2}{5}[BCE] = \frac{2}{5} \cdot \frac{45}{2} = 9$.
For $R$, use the base $AD$: $[CAD] = 15$ and $[FAD] = \frac{2}{3}[BAD] = 10$, so $CR : RF = 3 : 2$ and $[CAR] = \frac{3}{5}[CAF] = \frac{3}{5} \cdot 20 = 12$.
Therefore $[PQR] = 30 - 6 - 9 - 12 = 3$, one tenth of the triangle. The same three steps give the inner triangle for any three cevians; the general answer is Routh's theorem, which is not worth memorising for IMC or AMC 10.


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**Common traps.**

- Using Fact 1 for triangles whose bases are not on one line, or whose apex is not shared. Before writing a ratio, name the apex and the line.
- Writing $FP : PC$ as a ratio of the triangles on either side of $P$ without a common apex, for example $[BPF] : [CPE]$.
- In a trapezium, the triangles on the parallel sides are in the ratio $k^2$, and the two on the legs are equal. It is easy to use $k$ instead of $k^2$.
- Assuming that three cevians cut a triangle into pieces of equal area because the figure looks symmetric.
- Forgetting to draw the segment to the third vertex in Section 3: the quadrilateral $AFPE$ is not a triangle, and Fact 2 needs triangles.

### 1.4 How to spot this in a contest

1. The question gives only ratios on sides, or only areas of some regions, and asks for another area or ratio.
2. A figure with cevians meeting inside a triangle and numbers written in the regions.
3. A trapezium or parallelogram with its diagonals, or a segment from a vertex to the opposite side.
4. The answer is a fraction of the whole area: you may then fix the total area as any convenient number, such as the least common multiple of the denominators.
5. A coordinate solution needs you to choose a shape. If the answer cannot depend on the shape, look for an area argument first.

## 2 Practice problems

The problems are roughly in order of difficulty, from $\star$ to $\star\star\star$. Hints are in Section 4 and full solutions in Section 5.

**Problem 1** $\star$ Triangle $ABC$ has area 84. The point $D$ lies on $BC$ with $BD : DC = 3 : 4$, and $E$ lies on $AD$ with $AE : ED = 2 : 1$. Find the area of triangle $ABE$.



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**Problem 2** ⋆⋆ $ABCD$ is a trapezium with $AB \parallel DC$ and $AB = 2DC$. The point $M$ is the midpoint of $AB$, and $DM$ meets the diagonal $AC$ at $P$. What fraction of the area of $ABCD$ is the area of triangle $APD$?

**Problem 3** ⋆⋆ The cevians $AD$ and $BE$ of triangle $ABC$ meet at $P$. The areas of triangles $ABP$, $BPD$ and $APE$ are 12, 8 and 6. Find the area of quadrilateral $PDCE$.

**Problem 4** ⋆⋆⋆ Triangle $ABC$ has area 70. The point $D$ lies on $BC$ with $BD : DC = 1 : 3$, $E$ lies on $CA$ with $CE : EA = 2 : 3$, and $F$ is the midpoint of $AB$. The segments $AD$, $BE$, $CF$ bound a small triangle. Find its area.

**Problem 5** ⋆⋆⋆ Parallelogram $ABCD$ has area 140. The point $E$ lies on $AB$ with $AE : EB = 1 : 2$, and $F$ is the midpoint of $BC$. The segments $DE$ and $AF$ meet at $P$. Find the area of triangle $APD$.

**Problem 6** ⋆⋆⋆ $ABCD$ is a trapezium with $AB \parallel DC$ and $AB = 3DC$. Its diagonals meet at $X$, and the line through $X$ parallel to $AB$ meets $AD$ at $M$ and $BC$ at $N$. Find $\dfrac{[ABNM]}{[ABCD]}$.

## 3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **Purple Comet 2010 Middle School Problem 14** ⋆
   *Two of the four triangles at P are given: what is k, and what are the other two?*
2. **PMWC 2000 Problem T6** ⋆⋆
   *Work outwards from FGH: one application of Fact 1 for each ratio given, ending with apex A on line BC.*
3. **AMC 10A 2009 Problem 23** ⋆⋆
   *What do equal areas of AED and BEC say about the sides AB and CD?*
4. **AMC 8 2019 Problem 24** ⋆⋆
   *Label the small triangles and use ratios along BD and along BC.*
5. **NYSML Spring 1977 Problem 1** ⋆⋆
   *Fact 3 with a common base gives where each segment cuts the median.*
6. **AMC 10A 2016 Problem 19** ⋆⋆
   *Each of AE and AF forms a trapezium-type pair of similar triangles with the diagonal BD.*
7. **AMC 10B 2016 Problem 19** ⋆⋆⋆
   *Find where AC and AG cut EF from pairs of similar triangles with parallel sides.*
8. **AIME 1984 Problem 3** ⋆⋆⋆
   *The three corner triangles are similar to ABC; compare the square roots of the areas.*
9. **UKMT Senior Kangaroo 2024 Question 19** ⋆⋆⋆
   *The diagonal QT is parallel to RS and UP; use the trapezium ratios.*



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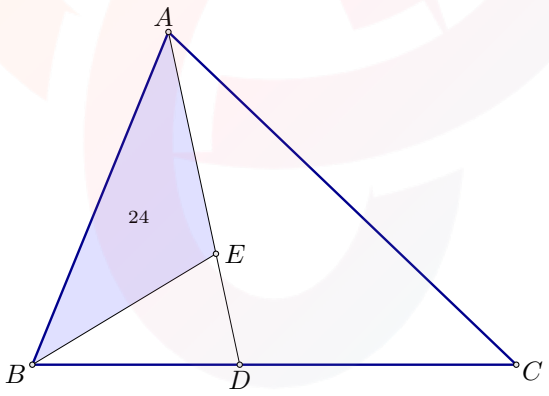
## 4 Hints

1. Two applications of Fact 1, first with apex $A$, then with apex $B$.
2. Compare $AM$ with $DC$. What kind of quadrilateral is $AMCD$?
3. Draw $CP$ and call the two new areas $x$ and $y$. Each of the cevians $AD$ and $BE$ gives one equation.
4. Remove three triangles, each with one side of $ABC$ as a side. For each one, Fact 3 gives the ratio along a cevian.
5. Extend $AF$ to meet line $DC$. You then have a trapezium whose diagonals meet at $P$.
6. Use Fact 5 to give the four triangles at $X$ areas in units. Then find where $M$ divides $AD$.

## 5 Solutions

### Problem 1

*Triangle $ABC$ has area 84. The point $D$ lies on $BC$ with $BD : DC = 3 : 4$, and $E$ lies on $AD$ with $AE : ED = 2 : 1$. Find the area of triangle $ABE$.*




Triangles $ABD$ and $ABC$ have apex $A$ and bases on $BC$, so $[ABD] = \frac{3}{7} \cdot 84 = 36$. Triangles $ABE$ and $ABD$ have apex $B$ and bases $AE$ and $AD$ on one line, so $[ABE] = \frac{2}{3} \cdot 36 = 24$.

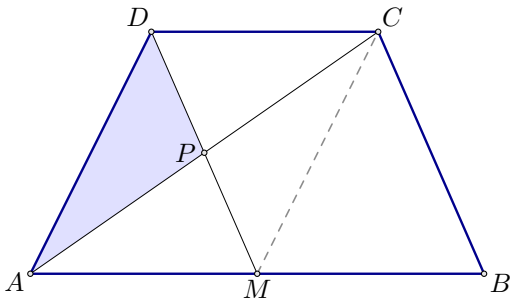
**Answer:** 24

### Problem 2

*$ABCD$ is a trapezium with $AB \parallel DC$ and $AB = 2DC$. The point $M$ is the midpoint of $AB$, and $DM$ meets the diagonal $AC$ at $P$. What fraction of the area of $ABCD$ is the area of triangle $APD$?*



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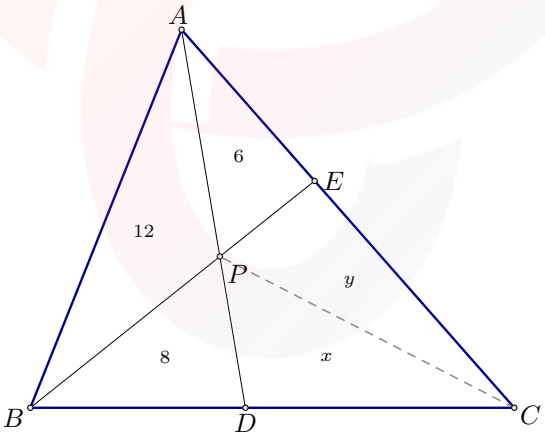



Let $T = [ACD]$. Triangles $ABC$ and $ACD$ have bases $AB$ and $DC$ on parallel lines and the same height, so $[ABC] = 2T$ and $[ABCD] = 3T$. Since $AM = DC$ and $AM \parallel DC$, the quadrilateral $AMCD$ is a parallelogram, and its diagonals $AC$ and $DM$ bisect each other at $P$. Its area is $[ACD] + [AMC] = T + \frac{1}{2}[ABC] = 2T$, and the diagonals cut it into four triangles of equal area, so $[APD] = \frac{T}{2}$. The fraction is $\frac{T/2}{3T} = \frac{1}{6}$.

**Answer:** $\dfrac{1}{6}$

### Problem 3

*The cevians $AD$ and $BE$ of triangle $ABC$ meet at $P$. The areas of triangles $ABP$, $BPD$ and $APE$ are 12, 8 and 6. Find the area of quadrilateral $PDCE$.*




Draw $CP$ and let $x = [PDC]$, $y = [PCE]$. Triangles $ABP$ and $BPD$ have apex $B$, so $AP : PD = 12 : 8 = 3 : 2$; with apex $C$ the same ratio gives $(6 + y) : x = 3 : 2$. Triangles $ABP$ and $APE$ have apex $A$, so $BP : PE = 12 : 6 = 2 : 1$; with apex $C$, $(8 + x) : y = 2 : 1$.

From the second equation $y = \frac{8+x}{2}$, and then the first gives $12 + (8 + x) = 3x$, so $x = 10$ and $y = 9$. The quadrilateral has area 19.

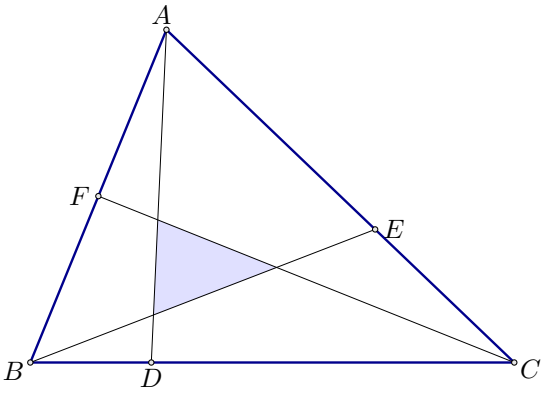
**Answer:** 19

### Problem 4

*Triangle $ABC$ has area 70. The point $D$ lies on $BC$ with $BD : DC = 1 : 3$, $E$ lies on $CA$ with $CE : EA = 2 : 3$, and $F$ is the midpoint of $AB$. The segments $AD$, $BE$, $CF$ bound a small triangle. Find its area.*



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Let $AD$ meet $BE$ at $P$, $BE$ meet $CF$ at $Q$, and $CF$ meet $AD$ at $R$. The small triangle is $PQR$, and $[PQR] = 70 - [ABP] - [BCQ] - [CAR]$.

By Fact 1, $[ABE] = \frac{3}{5} \cdot 70 = 42$, $[CBE] = 28$ and $[DBE] = \frac{1}{4} \cdot 28 = 7$. By Fact 3 with base $BE$, $AP : PD = 42 : 7 = 6 : 1$, so $[ABP] = \frac{6}{7}[ABD] = \frac{6}{7} \cdot \frac{70}{4} = 15$.

Next $[BCF] = [ACF] = 35$ and $[ECF] = \frac{2}{5} \cdot 35 = 14$, so $BQ : QE = 35 : 14 = 5 : 2$ and $[BCQ] = \frac{5}{7}[BCE] = \frac{5}{7} \cdot 28 = 20$.

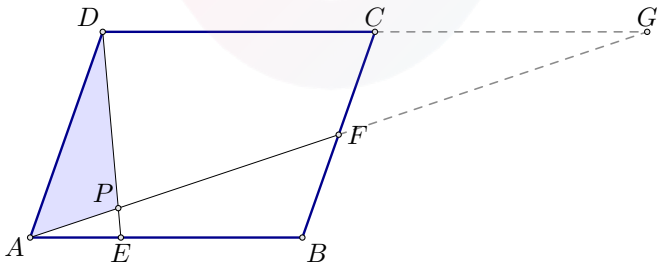
Finally $[CAD] = \frac{3}{4} \cdot 70 = \frac{105}{2}$ and $[FAD] = \frac{1}{2}[BAD] = \frac{35}{4}$, so $CR : RF = 6 : 1$ and $[CAR] = \frac{6}{7}[CAF] = \frac{6}{7} \cdot 35 = 30$.

Hence $[PQR] = 70 - 15 - 20 - 30 = 5$.

**Answer:** 5

### Problem 5

*Parallelogram $ABCD$ has area 140. The point $E$ lies on $AB$ with $AE : EB = 1 : 2$, and $F$ is the midpoint of $BC$. The segments $DE$ and $AF$ meet at $P$. Find the area of triangle $APD$.*




Extend $AF$ to meet line $DC$ at $G$. Triangles $ABF$ and $GCF$ have $BF = FC$, equal vertical angles at $F$ and equal alternate angles at $B$ and $C$, so they are congruent and $CG = AB$. Hence $DG = 2AB$.

Now $AE \parallel DG$, so $AEGD$ is a trapezium whose diagonals $AG$ and $DE$ meet at $P$, and $DG : AE = 2AB : \frac{1}{3}AB = 6$. The similar triangles $PDG$ and $PEA$ give $DP : PE = 6 : 1$. Since $[AED] = \frac{1}{3}[ABD] = \frac{1}{3} \cdot 70 = \frac{70}{3}$ (apex $D$, bases on $AB$), Fact 1 with apex $A$ gives $[APD] = \frac{6}{7} \cdot \frac{70}{3} = 20$.

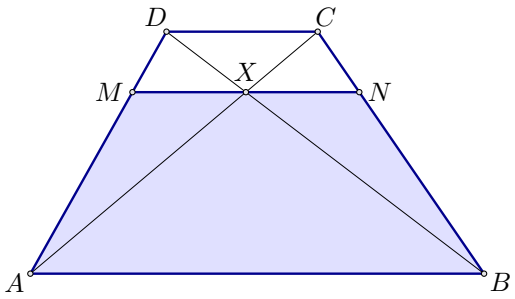
**Answer:** 20



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### Problem 6

*$ABCD$ is a trapezium with $AB \parallel DC$ and $AB = 3DC$. Its diagonals meet at $X$, and the line through $X$ parallel to $AB$ meets $AD$ at $M$ and $BC$ at $N$. Find $\dfrac{[ABNM]}{[ABCD]}$.*




By Fact 5 with $k = 3$, we may take $[CDX] = 1$, $[ABX] = 9$ and $[BCX] = [DAX] = 3$, so $[ABCD] = 16$. In triangle $DAB$ the line $MX$ is parallel to $AB$, so $DM : DA = DX : DB$. As $DX : XB = DC : AB = 1 : 3$, we get $DM : DA = 1 : 4$. Triangles $DMX$ and $DAX$ have apex $X$ and bases on line $DA$, so $[DMX] = \frac{1}{4} \cdot 3 = \frac{3}{4}$. In the same way $CN : CB = 1 : 4$ and $[CNX] = \frac{3}{4}$.

The part of the trapezium above $MN$ is made of $DMX$, $CDX$ and $CNX$, with total area $\frac{3}{4} + 1 + \frac{3}{4} = \frac{5}{2}$. Hence $[ABNM] = 16 - \frac{5}{2} = \frac{27}{2}$, and the ratio is $\frac{27}{32}$.

**Answer:** $\dfrac{27}{32}$